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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Regular and Supplementary Summer 2024</b><br><b>Course: B. Tech.      Branch : Mechanical Engineering      Semester : IV</b><br><b>Subject Code &amp; Name: BTMC402 Theory of Machines - I</b><br><b>Max Marks: 60                      Date: 14/06/2024                      Duration: 3 Hr.</b>             |                                                                                                                                                                                                                                                                                                                                     |            |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |                                                                                                                                                                                                                                                                                                                                     |            |           |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                     | (Level/CO) | Marks     |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                              |            | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Write types of ICR and explain Kennedy's theorem with example                                                                                                                                                                                                                                                                       | 1          | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | The radius of crank and length of connecting rod are 100 mm and 400 mm respectively. If the crank rotates clockwise with the angular velocity of 10 rad/s and is at 45 degree from inner dead center. Find velocity of slider using relative velocity Method.                                                                       | 2          | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | In a slider crank mechanism the radius of crank and length of connecting rod are 150 mm and 600 mm respectively. If the crank rotates clockwise with the angular velocity of 12 rad/s and is at 45 degree from inner dead center. Find velocity of slider and angular velocity of connecting rod using Klien's construction method. | 2          | <b>6</b>  |
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| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                              |            | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Write concept of friction axis                                                                                                                                                                                                                                                                                                      | 1          | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | In a thrust bearing the external and internal diameters of the contacting surfaces are 416 mm and 286 mm respectively. The total axial load is 125 kN and intensity of pressure 0.7 N/mm <sup>2</sup> . The shaft rotates at 370 RPM. Taking coefficient of friction 0.04 calculate power lost in overcoming the friction.          | 2          | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Derive an expression for frictional torque of flat collar bearing assuming uniform wear theory                                                                                                                                                                                                                                      | 2          | <b>6</b>  |
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| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                              |            | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | A multi plate clutch has 6 active friction surfaces. Power transmitted is 20KW at 400 RPM. Inner and outer radii of friction surface are 90mm and                                                                                                                                                                                   | 2          | <b>6</b>  |

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|                                             | 120mm respectively. Assuming uniform wear and coefficient of friction 0.3. Find the maximum intensity of pressure between discs.                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |           |
| B)                                          | Derive an expression for torque transmitted by cone clutch due to friction considering uniform pressure theory                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2 | 6         |
| C)                                          | Determine the axial force required to engage cone clutch transmitting 25 KW power at 250 RPM. Average friction diameter of cone is 400 mm, semi cone angle $10^\circ$ and coefficient of friction 0.25                                                                                                                                                                                                                                                                                                                                                            | 2 | 6         |
|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |           |
| <b>Q.4 Solve Any Two of the following.</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   | <b>12</b> |
| A)                                          | Derive an expression for braking torque for single block brake when the line of action of tangential braking force is at distance 'a' below the fulcrum 'O' at clockwise rotation of drum.                                                                                                                                                                                                                                                                                                                                                                        | 2 | 6         |
| B)                                          | In a rope brake dynamometer Diameter of flywheel 1m, Diameter of rope 10mm, Dead weight of the brake 50Kg, Speed of Engine 180 rpm, spring balance reading 120 N, Find brake power of the Engine                                                                                                                                                                                                                                                                                                                                                                  | 3 | 6         |
| C)                                          | A torsion dynamometer is fitted on turbine shaft to measure the angle of twists. It is observed that the angle of twist. It observed that the shaft twists in $1.5^\circ$ in a length of 5m at 500 rpm. The shaft is solid and has diameter of 200 mm. Modulus of rigidity of shaft material is 85 GPa. Find the power transmitted by turbine.                                                                                                                                                                                                                    | 3 | 6         |
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| <b>Q. 5 Solve Any Two of the following.</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   | <b>12</b> |
| A)                                          | Explain with neat sketch types of cam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 | 6         |
| B)                                          | A cam rotating at uniform speed required to give the Knife edge follower motions as:<br>(i) Follower to move outwards through the distance of 30mm during $90^\circ$ of cam rotation with uniform velocity<br>(ii) Follower to dwell for next $60^\circ$ of cam rotation<br>(iii) Follower to return to its original position during $90^\circ$ cam rotation with uniform velocity<br>(iv) Follower to dwell for remaining period.<br>The minimum radius of the cam is 50 mm. Draw the cam profile when follower is offset by 20mm from the axis of the cam shaft | 3 | 6         |
| C)                                          | Four masses $m_1$ , $m_2$ , $m_3$ and $m_4$ are 200 kg, 300 kg, 240 kg and 260 kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 | 6         |

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|  | respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are $45^\circ$ , $75^\circ$ and $135^\circ$ . Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m |  |  |
|  | *** End ***                                                                                                                                                                                                                                                                                |  |  |